

Week 2: Reading, Exporting, and Checking Data

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2025-09-10

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Chapter 1

Overview

We will cover importing and exporting data. Some of this may be a review, but there are helpful options that you may have missed. We will also cover ways to check for data quality issues. This will be an important session, especially considering measurement error is a potential source of bias.

Note: Recommended resources UCLA OARC Stata resources: <https://stats.oarc.ucla.edu/stata/modules/>

I have used this source on numerous times. Even when you become proficient with Stata, UCLA's OARC Stata resources have been very helpful. If you just type *UCLA OARC Stata* in a search engine, then it will usually pop up.

Statalist is usually another good resource when trying to figure out what went wrong. If you ask a question on the listserve, you will likely be scolded by Nick Cox, or you will be told to "read the manual". However, if you just read what others have asked, you will often find useful information.

Overview: - Importing Data - Exporting Data - Data Checks and Cleaning

Chapter 2

Ingesting Data

Mitchell Chapter 2

Note when your working directory, I recommend using “/” instead of “” Stata will use both, and the backslash”” often runs into string issues.

2.1 Getting our data

We’ll get our data that we will use for the course.

```
pwd
*Set your working directory
cd "/Users/Sam/Desktop/Econ 645/Data/"

*Use:
net from https://www.stata-press.com/data/dmus2
net get dmus1
net get dmus2

*All commands:
net install dmus1
```

2.2 Reading Stata data

We use the *use* command to read Stata data. Use the *clear* option to clear the data from memory so you can read the data file. Otherwise, you will get an error

We’ll add the *clear* option to clear our memory to provide an error

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
use "dentists.dta", clear
list
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```

+-----+
|              name    years    fulltime    recom |
+-----+
1. | Y. Don Uflossmore    7.25         0         1 |
2. |   Olive Tu'Drill    10.25         1         1 |
3. | Isaac O'Yerbreath    32.75         1         1 |
4. |   Ruth Canaale      22         1         1 |
5. |   Mike Avity        8.5         0         0 |
+-----+
```

We can read Stata data from websources. From Stata Press website:

```
use "https://www.stata-press.com/data/dmus2/dentists.dta", clear
list
```

```

|              name    years    fulltime    recom |
+-----+
1. | Y. Don Uflossmore    7.25         0         1 |
2. |   Olive Tu'Drill    10.25         1         1 |
3. | Isaac O'Yerbreath    32.75         1         1 |
4. |   Ruth Canaale      22         1         1 |
5. |   Mike Avity        8.5         0         0 |
+-----+
```

From NBER website

```
use "https://data.nber.org/morg/annual/morg20.dta", clear
list lfsr94 stfips sex in f/10
```

```

|              lfsr94    stfips    sex |
+-----+
1. |      Employed-At Work      AL      2 |
2. |      Employed-At Work      AL      1 |
3. | Retired-Not In Labor Force      AL      1 |
4. | Retired-Not In Labor Force      AL      2 |
5. |   Other-Not In Labor Force      AL      2 |
+-----+
```


6.		Disabled-Not In Labor Force	AL	1	
7.		Retired-Not In Labor Force	AL	1	
8.		Retired-Not In Labor Force	AL	2	
9.		Retired-Not In Labor Force	AL	2	
10.		Retired-Not In Labor Force	AL	1	

+-----+

2.3 Reading Stata data with subsets

Subsetting can be a helpful method to read in extremely large data sets, such as American Community Survey (ACS) public use file. We can subset by variables, observations, or both.

2.3.1 Subsetting by columns (variables)

You can read in a subset of Stata data by columns. This can be helpful when you have a large dataset but only need a few columns of data. However, you need to know the variable names. We will need to have a data dictionary in order to do this. The January CPS data dictionary can be found on Census's website. https://www2.census.gov/programs-surveys/cps/datasets/2025/basic/2025_Basic_CPS_Public_Use_Record_Layout_plus_IO_Code_list.txt

```
quietly cd "/Users/Sam/Desktop/Data/CPS/"
use gestfips pemlr pternwa using "smalljul25pub.dta", clear
list in 1/3
```

		gestfips	pemlr	pternwa	

1.		1	.	.	
2.		1	5	-1	
3.		1	6	-1	

+-----+

If it is a large dataset, I don't recommend using list by itself. Use in 1/N

2.3.2 Subsetting by rows (observations)

You can read in a subset of data by rows. In my opinion, this is a better method to subset than by columns. You are able to see the entire dataset. You can use a qualifiers *if* or *in*

```
quietly cd "/Users/Sam/Desktop/Data/CPS/"
use "smalljul25pub.dta" if gestfip == 24 & prtage == 30, clear
list in 1/10
```

```
1. | hrhhid2 | hrmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puic
   | 17011 | 7 | 7 | 2025 | 24 | 5 | 3 | -1 |
   +-----+-----+-----+-----+-----+-----+-----+-----+
   | perrp | pesex | peparitl | pehspon | peeduca | peafever | ptdtrace | prperty
   | 49 | 1 | 6 | 2 | 40 | 2 | 2 |
   +-----+-----+-----+-----+-----+-----+-----+
   | pehract1 | pehract2 | pemlr | peio1cd | peio1cow | peio2cow | prmjind1 | prmjind2
   | 12 | -1 | 1 | 6380 | 4 | -1 | 6 |
   +-----+-----+-----+-----+-----+-----+-----+
   | prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernhri
   | 23 | -1 | 17 | -1 | 1 | -1 | -1 |
   +-----+-----+-----+-----+-----+-----+-----+
   | pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1cd | hrhhid2
   | 2 | -1 | -1 | 99149562 | -1 | 5510 | 9.401e+
   +-----+-----+-----+-----+-----+-----+-----+

2. | hrhhid2 | hrmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puic
   | 17011 | 6 | 7 | 2025 | 24 | 5 | 2 | -1 |
   +-----+-----+-----+-----+-----+-----+-----+
   | perrp | pesex | peparitl | pehspon | peeduca | peafever | ptdtrace | prperty
   | 51 | 1 | 1 | 2 | 39 | 2 | 4 |
   +-----+-----+-----+-----+-----+-----+-----+
   | pehract1 | pehract2 | pemlr | peio1cd | peio1cow | peio2cow | prmjind1 | prmjind2
   | 44 | 8 | 1 | 770 | 4 | -1 | 3 |
   +-----+-----+-----+-----+-----+-----+-----+
   | prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernhri
   | 4 | -1 | 20 | -1 | 2 | -1 | -1 |
   +-----+-----+-----+-----+-----+-----+-----+
   | pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1cd | hrhhid2
   | 2 | -1 | -1 | 48285071 | -1 | 7315 | 4.700e+
   +-----+-----+-----+-----+-----+-----+-----+

3. | hrhhid2 | hrmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puic
   | 17012 | 8 | 7 | 2025 | 24 | 5 | 1 | -1 |
   +-----+-----+-----+-----+-----+-----+-----+
   | perrp | pesex | peparitl | pehspon | peeduca | peafever | ptdtrace | prperty
   | 41 | 1 | 6 | 2 | 42 | 2 | 2 |
   +-----+-----+-----+-----+-----+-----+-----+
   | pehract1 | pehract2 | pemlr | peio1cd | peio1cow | peio2cow | prmjind1 | prmjind2
```

	40	-1	1	7280	4	-1	9	-1	
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peernlab	pre
	36	-1	17	-1	1	2	-1	2	
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhhid	ptwk	
	2	-1	-1	113138434	183000	5120	3.411e+14	0	

4.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis	puiodp1	puio
	19011	3	7	2025	24	5	4	-1	-1	
	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prpertyp	prtag	
	49	1	6	2	39	2	1	2	3	
	pehract1	pehract2	pemlr	peiolcd	peio1cow	peio2cow	prmjind1	prmjind2	prmjc	
	32	-1	1	770	4	-1	3	-1		
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peernlab	pre	
	4	-1	22	-1	1	-1	-1	-1		
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhhid	ptwk		
	2	-1	-1	44333596	-1	9130	9.007e+14	0		

5.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis	puiodp1	puio
	19011	2	7	2025	24	5	1	-1	1	
	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prpertyp	prtag	
	40	1	1	2	39	2	1	2	3	
	pehract1	pehract2	pemlr	peiolcd	peio1cow	peio2cow	prmjind1	prmjind2	prmjc	
	50	-1	1	170	4	-1	1	-1		
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peernlab	pre	
	1	-1	18	-1	1	-1	-1	-1		
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhhid	ptwk		
	2	-1	-1	31435370	-1	6050	9.450e+14	0		

6.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis	puiodp1	puio
	17011	5	7	2025	24	5	1	-1	-1	

	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
	41	1	6	1	45	2	4	
	pehract1	pehract2	pemlr	peiolicd	peio1cow	peio2cow	prmjind1	prmjin
	40	-1	1	7980	4	-1	10	
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
	42	-1	10	-1	1	-1	-1	
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh	
	1	1	1	75007444	-1	3010	4.260e+	
7.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis
	19011	3	7	2025	24	5	2	1
	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
	58	1	6	2	39	2	2	
	pehract1	pehract2	pemlr	peiolicd	peio1cow	peio2cow	prmjind1	prmjin
	-1	-1	6	-1	-1	-1	-1	
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
	-1	-1	-1	-1	-1	-1	-1	
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh	
	2	-1	-1	106522944	-1	-1	6.604e+	
8.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis
	17011	6	7	2025	24	5	1	-1
	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
	40	1	6	2	43	2	1	
	pehract1	pehract2	pemlr	peiolicd	peio1cow	peio2cow	prmjind1	prmjin
	43	-1	1	6481	4	-1	7	
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
	25	-1	16	-1	1	-1	-1	
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh	
	2	-1	-1	34698967	-1	4850	4.223e+	

```

+-----+
+-----+
9. | hrhhid2 | hrmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio
   | 19011 | 3 | 7 | 2025 | 24 | 5 | 1 | -1 | 1 |
+-----+
| perrp | pesex | pemaritl | pehspon | peeduca | peafever | ptdtrace | prpertyp | prtag
   | 41 | 1 | 6 | 1 | 43 | 2 | 1 | 2 | 3 |
+-----+
| pehract1 | pehract2 | pemlr | peiolcd | peio1cow | peio2cow | prmjind1 | prmjind2 | prmj
   | 50 | -1 | 1 | 7071 | 4 | -1 | 8 | -1 |
+-----+
| prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | pre
   | 34 | -1 | 2 | -1 | 1 | -1 | -1 | -1 |
+-----+
| pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1ocd | hrhhid | ptwk
   | 2 | -1 | -1 | 50143408 | -1 | 800 | 1.913e+14 | 0 |
+-----+

+-----+
+-----+
10. | hrhhid2 | hrmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio
    | 17011 | 5 | 7 | 2025 | 24 | 5 | 1 | -1 | -1 |
+-----+
| perrp | pesex | pemaritl | pehspon | peeduca | peafever | ptdtrace | prpertyp | prtag
    | 40 | 2 | 1 | 2 | 44 | 2 | 1 | 2 | 3 |
+-----+
| pehract1 | pehract2 | pemlr | peiolcd | peio1cow | peio2cow | prmjind1 | prmjind2 | prmj
    | 5 | -1 | 1 | 7460 | 5 | -1 | 9 | -1 |
+-----+
| prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | pre
    | 36 | -1 | 3 | -1 | 1 | -1 | -1 | -1 |
+-----+
| pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1ocd | hrhhid | ptwk
    | 2 | -1 | -1 | 87833706 | -1 | 1006 | 2.911e+14 | 0 |
+-----+

```

```

quietly cd "/Users/Sam/Desktop/Data/CPS/"
use "smalljul25pub.dta" in 1/10, clear
list

```

```

1. | hrhhid2 | hrmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio
   | 17011 | 8 | 7 | 2025 | 1 | 6 | . | . | . |
+-----+
| perrp | pesex | pemaritl | pehspon | peeduca | peafever | ptdtrace | prpertyp | prtag
   | . | . | . | . | . | . | . | . |

```

	pehract1	pehract2	pemlr	peiolicd	peio1cow	peio2cow	prmjind1	prmjin
	.	.	.	.	.	.	.	.
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
	.	.	.	.	.	.	.	.
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh	
	.	.	.	.	.	.	1.617e+	
2.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis
	17011	8	7	2025	1	6	1	-1
	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
	40	2	1	2	43	2	1	
	pehract1	pehract2	pemlr	peiolicd	peio1cow	peio2cow	prmjind1	prmjin
	-1	-1	5	-1	-1	-1	-1	
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
	-1	-1	-1	-1	-1	-1	-1	
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh	
	2	-1	-1	18467177	-1	-1	1.090e+	
3.	hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis
	17011	8	7	2025	1	6	2	1
	perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
	42	1	1	2	43	1	1	
	pehract1	pehract2	pemlr	peiolicd	peio1cow	peio2cow	prmjind1	prmjin
	-1	-1	6	-1	-1	-1	-1	
	prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
	-1	-1	-1	-1	-1	-1	-1	
	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh	
	2	-1	-1	27264237	-1	-1	1.090e+	

```

4. | hrhhid2 | hrmmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio
   | 17011 | 8 | 7 | 2025 | 1 | 6 | 1 | -1 | -1 |
   +-----+-----+-----+
   | perrp | pesex | pamaritl | pehspon | peeduca | peafever | ptdtrace | prperty | prtag
   | 40 | 1 | 1 | 2 | 43 | 1 | 1 | 2 | 6
   +-----+-----+-----+
   | pehract1 | pehract2 | pemlr | peiolcd | peio1cow | peio2cow | prmjind1 | prmjind2 | prmjc
   | -1 | -1 | 5 | -1 | -1 | -1 | -1 | -1 |
   +-----+-----+-----+
   | prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | pre
   | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
   +-----+-----+-----+
   | pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1ocd | hrhhid | ptwk
   | 2 | -1 | -1 | 24488360 | -1 | -1 | 6.101e+14 | 0
   +-----+-----+-----+

5. | hrhhid2 | hrmmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio
   | 17011 | 8 | 7 | 2025 | 1 | 6 | 2 | -1 | 1 |
   +-----+-----+-----+
   | perrp | pesex | pamaritl | pehspon | peeduca | peafever | ptdtrace | prperty | prtag
   | 42 | 2 | 1 | 2 | 39 | 2 | 1 | 2 | 6
   +-----+-----+-----+
   | pehract1 | pehract2 | pemlr | peiolcd | peio1cow | peio2cow | prmjind1 | prmjind2 | prmjc
   | 4 | -1 | 1 | 7390 | 4 | -1 | 9 | -1 |
   +-----+-----+-----+
   | prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | pre
   | 36 | -1 | 2 | -1 | 1 | 1 | 4 | 2 |
   +-----+-----+-----+
   | pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1ocd | hrhhid | ptwk
   | 2 | -1 | -1 | 18946917 | 8800 | 710 | 6.101e+14 | 0
   +-----+-----+-----+

6. | hrhhid2 | hrmmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio
   | 17011 | 7 | 7 | 2025 | 1 | 6 | 1 | -1 | -1 |
   +-----+-----+-----+
   | perrp | pesex | pamaritl | pehspon | peeduca | peafever | ptdtrace | prperty | prtag
   | 41 | 2 | 3 | 2 | 36 | 2 | 1 | 2 | 6
   +-----+-----+-----+
   | pehract1 | pehract2 | pemlr | peiolcd | peio1cow | peio2cow | prmjind1 | prmjind2 | prmjc
   | -1 | -1 | 5 | -1 | -1 | -1 | -1 | -1 |
   +-----+-----+-----+
   | prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | pre
   | -1 | -1 | -1 | -1 | -1 | -1 | -1 | -1 |
   +-----+-----+-----+

```

	pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh
	2	-1	-1	20570322	-1	-1	9.101e+

7.

hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis	puic
17011	8	7	2025	1	6	.	.	.

perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
.	.	.	.	.	.	.	.

pehract1	pehract2	pemlr	peio1icd	peio1cow	peio2cow	prmjind1	prmjin
.	.	.	.	.	.	.	.

prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
.	.	.	.	.	.	.	.

pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh
.	.	.	.	.	.	4.524e+

8.

hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis	puic
17011	8	7	2025	1	6	.	.	.

perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
.	.	.	.	.	.	.	.

pehract1	pehract2	pemlr	peio1icd	peio1cow	peio2cow	prmjind1	prmjin
.	.	.	.	.	.	.	.

prdtind1	prdtind2	prdtocc1	prdtocc2	prsjmj	peernhry	peernhro	peern
.	.	.	.	.	.	.	.

pecert1	pecert2	pecert3	pwcmpwgt	pternwa	ptio1ocd	hrhh
.	.	.	.	.	.	1.101e+

9.

hrhhid2	hrmis	hrmonth	hryear4	gestfips	gediv	pulineno	pudis	puic
17011	8	7	2025	1	6	1	-1	.

perrp	pesex	pemaritl	pehspnon	peeduca	peafever	ptdtrace	prperty
40	2	1	2	44	2	1	.

pehract1	pehract2	pemlr	peio1icd	peio1cow	peio2cow	prmjind1	prmjin
----------	----------	-------	----------	----------	----------	----------	--------


```

|          -1 |          -1 |          5 |          -1 |          -1 |          -1 |          -1 |          -1 |
|-----+-----+-----+-----+-----+-----+-----+-----+-----+
| prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | prerelg |
|          -1 |          -1 |          -1 |          -1 |          -1 |          -1 |          -1 |          -1 |
|-----+-----+-----+-----+-----+-----+-----+-----+
| pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1ocd |          hrhhid |          ptwks |
|          2 |          -1 |          -1 | 19775533 |          -1 |          -1 | 5.314e+14 |          0 |
|-----+-----+-----+-----+-----+-----+-----+-----+
10. | hrhhid2 | hrmmis | hrmonth | hryear4 | gestfips | gediv | pulineno | pudis | puiodp1 | puio1ocd |
|      17011 |          8 |          7 |      2025 |          1 |          6 |          2 |          -1 |          -1 |
|-----+-----+-----+-----+-----+-----+-----+-----+
| perrp | pesex | peparitl | pehspon | peeduca | peafever | ptdtrace | prperty | prtag |
|          42 |          1 |          1 |          2 |          43 |          2 |          1 |          2 |          6 |
|-----+-----+-----+-----+-----+-----+-----+-----+
| pehract1 | pehract2 | pemlr | peio1cd | peio1cow | peio2cow | prmjind1 | prmjind2 | prmjind3 |
|          -1 |          -1 |          5 |          -1 |          -1 |          -1 |          -1 |          -1 |
|-----+-----+-----+-----+-----+-----+-----+-----+
| prdtind1 | prdtind2 | prdtocc1 | prdtocc2 | prsjmj | peernhry | peernhro | peernlab | prerelg |
|          -1 |          -1 |          -1 |          -1 |          -1 |          -1 |          -1 |          -1 |
|-----+-----+-----+-----+-----+-----+-----+-----+
| pecert1 | pecert2 | pecert3 | pwcmpwgt | pternwa | ptio1ocd |          hrhhid |          ptwks |
|          2 |          -1 |          -1 | 24488360 |          -1 |          -1 | 5.314e+14 |          0 |
|-----+-----+-----+-----+-----+-----+-----+-----+

```

Subsetting by row and columns

```

quietly cd "/Users/Sam/Desktop/Data/CPS/"
use gestfips pemlr pternwa prerelg using "smalljul25pub.dta" if gestfip == 24 & prerelg==1, clear
list in 60/69

```

```

| gestfips | pemlr | prerelg | pternwa |
|-----+-----+-----+-----+
60. |      24 |      1 |      1 | 132000 |
61. |      24 |      1 |      1 |  84000 |
62. |      24 |      1 |      1 | 108000 |
63. |      24 |      1 |      1 |  80000 |
64. |      24 |      1 |      1 | 163000 |
|-----+-----+-----+-----+
65. |      24 |      1 |      1 | 170000 |
66. |      24 |      1 |      1 |  70000 |
67. |      24 |      1 |      1 |  54000 |
68. |      24 |      1 |      1 | 252000 |
69. |      24 |      1 |      1 |  60000 |
|-----+-----+-----+-----+

```

2.3.3 System data

There are data sets that are included in Stata and use the *sysuse* command.

```
sysuse auto, clear
list make price mpg rep78 foreign, sep(5)
```

(1978 Automobile Data)

	make	price	mpg	rep78	foreign
1.	AMC Concord	4,099	22	3	Domestic
2.	AMC Pacer	4,749	17	3	Domestic
3.	AMC Spirit	3,799	22	.	Domestic
4.	Buick Century	4,816	20	3	Domestic
5.	Buick Electra	7,827	15	4	Domestic
6.	Buick LeSabre	5,788	18	3	Domestic
7.	Buick Opel	4,453	26	.	Domestic
8.	Buick Regal	5,189	20	3	Domestic
9.	Buick Riviera	10,372	16	3	Domestic
10.	Buick Skylark	4,082	19	3	Domestic
11.	Cad. Deville	11,385	14	3	Domestic
12.	Cad. Eldorado	14,500	14	2	Domestic
13.	Cad. Seville	15,906	21	3	Domestic
14.	Chev. Chevette	3,299	29	3	Domestic
15.	Chev. Impala	5,705	16	4	Domestic
16.	Chev. Malibu	4,504	22	3	Domestic
17.	Chev. Monte Carlo	5,104	22	2	Domestic
18.	Chev. Monza	3,667	24	2	Domestic
19.	Chev. Nova	3,955	19	3	Domestic
20.	Dodge Colt	3,984	30	5	Domestic
21.	Dodge Diplomat	4,010	18	2	Domestic
22.	Dodge Magnum	5,886	16	2	Domestic
23.	Dodge St. Regis	6,342	17	2	Domestic
24.	Ford Fiesta	4,389	28	4	Domestic
25.	Ford Mustang	4,187	21	3	Domestic
26.	Linc. Continental	11,497	12	3	Domestic
27.	Linc. Mark V	13,594	12	3	Domestic
28.	Linc. Versailles	13,466	14	3	Domestic

29.	Merc. Bobcat	3,829	22	4	Domestic	
30.	Merc. Cougar	5,379	14	4	Domestic	

31.	Merc. Marquis	6,165	15	3	Domestic	
32.	Merc. Monarch	4,516	18	3	Domestic	
33.	Merc. XR-7	6,303	14	4	Domestic	
34.	Merc. Zephyr	3,291	20	3	Domestic	
35.	Olds 98	8,814	21	4	Domestic	

36.	Olds Cutl Supr	5,172	19	3	Domestic	
37.	Olds Cutlass	4,733	19	3	Domestic	
38.	Olds Delta 88	4,890	18	4	Domestic	
39.	Olds Omega	4,181	19	3	Domestic	
40.	Olds Starfire	4,195	24	1	Domestic	

41.	Olds Toronado	10,371	16	3	Domestic	
42.	Plym. Arrow	4,647	28	3	Domestic	
43.	Plym. Champ	4,425	34	5	Domestic	
44.	Plym. Horizon	4,482	25	3	Domestic	
45.	Plym. Sapporo	6,486	26	.	Domestic	

46.	Plym. Volare	4,060	18	2	Domestic	
47.	Pont. Catalina	5,798	18	4	Domestic	
48.	Pont. Firebird	4,934	18	1	Domestic	
49.	Pont. Grand Prix	5,222	19	3	Domestic	
50.	Pont. Le Mans	4,723	19	3	Domestic	

51.	Pont. Phoenix	4,424	19	.	Domestic	
52.	Pont. Sunbird	4,172	24	2	Domestic	
53.	Audi 5000	9,690	17	5	Foreign	
54.	Audi Fox	6,295	23	3	Foreign	
55.	BMW 320i	9,735	25	4	Foreign	

56.	Datsun 200	6,229	23	4	Foreign	
57.	Datsun 210	4,589	35	5	Foreign	
58.	Datsun 510	5,079	24	4	Foreign	
59.	Datsun 810	8,129	21	4	Foreign	
60.	Fiat Strada	4,296	21	3	Foreign	

61.	Honda Accord	5,799	25	5	Foreign	
62.	Honda Civic	4,499	28	4	Foreign	
63.	Mazda GLC	3,995	30	4	Foreign	
64.	Peugeot 604	12,990	14	.	Foreign	
65.	Renault Le Car	3,895	26	3	Foreign	

66.	Subaru	3,798	35	5	Foreign	

67.	Toyota Celica	5,899	18	5	Foreign	
68.	Toyota Corolla	3,748	31	5	Foreign	
69.	Toyota Corona	5,719	18	5	Foreign	
70.	VW Dasher	7,140	23	4	Foreign	

71.	VW Diesel	5,397	41	5	Foreign	
72.	VW Rabbit	4,697	25	4	Foreign	
73.	VW Scirocco	6,850	25	4	Foreign	
74.	Volvo 260	11,995	17	5	Foreign	
	+-----+					

You can use Stata files that are available, but didn't ship with Stata and they use the *webuse* command

```
webuse fullauto, clear
list make price mpg rep78 foreign in 1/5
```

(Automobile Models)

		make	price	mpg	rep78	foreign	

1.		AMC	4099	22	Average	Domestic	
2.		AMC	4749	17	Average	Domestic	
3.		AMC	3799	22	.	Domestic	
4.		Audi	6295	23	Average	Foreign	
5.		Audi	9690	17	Excellent	Foreign	

2.4 Importing Excel files

Many times you will need to grab data from a website and these data may be in an Excel format such as ".xls" or ".xlsx". Our main command is *import excel*. We need to specify the *firstrow* option if we want the headers as variable names. Be careful if the headers are numerics, such as years, in the firstrow. This will throw an error and we'll need to rename them in Excel before importing these files. For example, 2015 becomes y2015.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
import excel using "dentists.xls", firstrow clear
list
```

	name	years	fulltime	recom	
--	------	-------	----------	-------	--

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

We need to take caution of the sheet names of the excel files we are importing. If you don't pay attention you may import the wrong sheet.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
import excel using "dentists2.xls", firstrow clear
list

import excel using "dentists2.xls", firstrow clear sheet("dentists")
list
```

	name	years
1.	I. Sue Yoo	3
2.	A. Dewey	8
3.	B. Cheetem	11
4.	C. Howe	21

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

2.5 Importing SAS, SPSS, dbase files

In my opinion, importing files from other statistical packages is a pain. If you are using another statistical software packages like R, SAS, Python, or heaven-forbid SPSS, then export the data from those software package as CSV files. CSV files are fairly universal. There maybe some labeling issues, but if labels are not a concern then use CSV file formats.

Luckily, SAS data files are becoming less common. Even the Census Bureau has seen the light and provides their data in a csv format.

2.5.1 Importing .sas7bdat

For me, it is unavailable in Stata 14, but if you have Stata 16 or higher you can use this command.

Import sas is our main command to import SAS from version 7 or higher.

```
import sas "dentists.sas7bdat", clear  
list
```

2.5.2 Importing SAS XPORT Version 5

```
import sasxport5 "dentists.xpt", clear  
list
```

2.5.3 Importing SAS XPORT Version 8

```
import sasxport8 "dentists.xpt", clear  
list
```

2.5.4 Importing SPSS files

For me, it is unavailable in Stata 14, but if you have Stata 16 or higher you can use this command.

```
import spss "dentlab.sav", clear  
list
```

2.5.5 Importing dBase files

For me, it is unavailable in Stata 14, but if you have Stata 16 or higher you can use this command.

```
import dbase "dentlab.dbf", clear  
list
```

2.6 Importing Raw Data files

We use the *import delimited* command to read in csv, txt, dat, etc. files.

2.6.1 Import Delimited

Importing csv files will likely be more common than other types of files so far with the exception of Stata and Excel files. *Note:* there are other types of delimited files, but comma-delimited are the most common. Other types of delimiters include “tab”, “space”, or “|” delimited.

CSV file If you read the manual you’ll noticed that “,” and “^” are the default delimiters Using “,” delimiter:

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists.csv", clear
list

quietly cd "/Users/Sam/Desktop/Data/CPS/"
import delimited using "jul25pub.csv", clear
list gestfips pmlr pternwa in 1/10
```

(4 vars, 5 obs)

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

(393 vars, 122,252 obs)

	gestfips	pmlr	pternwa
1.	1	.	.
2.	1	5	-1
3.	1	6	-1
4.	1	5	-1

```

5. |      1      1      8800 |
   |-----|
6. |      1      5      -1 |
7. |      1      .      . |
8. |      1      .      . |
9. |      1      5      -1 |
10. |     1      5      -1 |
   +-----+

```

CSV files may come in .txt files or .csv files using “,” delimiter

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists1.txt", clear
list

```

(4 vars, 5 obs)

```

+-----+
|      name      years  fulltime  recom |
+-----+
1. | Y. Don Uflossmore    7.25        0      1 |
2. |   Olive Tu'Drill    10.25        1      1 |
3. | Isaac O'Yerbreath   32.75        1      1 |
4. |   Ruth Canaale      22          1      1 |
5. |   Mike Avity       8.5          0      0 |
+-----+

```

There may be “tab”-delimited files that use “`^`” delimiter

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists2.txt", clear
list

```

(4 vars, 5 obs)

```

+-----+
|      name      years  fulltime  recom |
+-----+
1. | Y. Don Uflossmore    7.25        0      1 |
2. |   Olive Tu'Drill    10.25        1      1 |
3. | Isaac O'Yerbreath   32.75        1      1 |
4. |   Ruth Canaale      22          1      1 |
5. |   Mike Avity       8.5          0      0 |
+-----+

```


This will produce the same results as above since “” is a default

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists2.txt", clear delimiters("\t")
list
```

(4 vars, 5 obs)

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

If we set the delimiter to “:”, we can import a “:”-delimited file

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists4.txt", clear delimiters(":")
list
```

(4 vars, 5 obs)

	v1	v2	v3	v4
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

We can subset columns and rows with import delimited, which can be helpful for large files. Variable names will be defaulted for the first few rows, so when we use rowrange(1:3) this will only include 2 rows of data.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists2.txt", clear rowrange(1:3)
list
```

(4 vars, 2 obs)

```

+-----+
|           name    years   fulltime   recom |
+-----+
1. | Y. Don Uflossmore    7.25         0         1 |
2. |   Olive Tu'Drill    10.25         1         1 |
+-----+

```

Import delimited from the web

```

import delimited using "https://www2.census.gov/programs-surveys/cps/datasets/2023/bas
pwd
save "jun23pub.dta", replace

```

2.6.2 Import delimited with subset of rows

You can use the *rowrange* option to subset large dataset. For example, the NPPES admin data is over 10 GB. If we want to inspect the data, we can just use *rowrange*. Another example would be the American Community Survey Public Use File.

```

cd "/Users/Sam/Desktop/Data/NPPES/"
import delimited using npidata_pfile_20050523-20241208.csv, varnames(1) rowrange(1:10)

```

2.6.3 Import space-separated files

Sometimes files have space-delimited files (which is a bad idea when strings are involved). *Dentists5.txt* is an example of such a file. We can use the *infile* command to import these data, but we need to specify that the name variable is a string at least 17 characters wide.

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
infile str17 name years full rec using "dentists5.txt", clear
list

```

(5 observations read)

```

+-----+
|           name    years   full   rec |
+-----+
1. | Y. Don Uflossmore    7.25         0     1 |

```

```

2. |      Olive Tu'Drill    10.25      1      1 |
3. |   Isaac O'Yerbreath   32.75      1      1 |
4. |      Ruth Canaale      22        1      1 |
5. |      Mike Avity        8.5        0      0 |
+-----+

```

Infile does not read in variable names in the first row.

We can also use `import delimited` with `delimiters(" ")`

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
import delimited using "dentists6.txt", delimiters(" ") clear
list

```

(4 vars, 5 obs)

```

+-----+
|              name    years    fulltime    recom |
+-----+
1. | Y. Don Uflossmore    7.25        0        1 |
2. |   Olive Tu'Drill    10.25        1        1 |
3. | Isaac O'Yerbreath   32.75        1        1 |
4. |      Ruth Canaale    22          1        1 |
5. |      Mike Avity      8.5          0        0 |
+-----+

```

2.6.4 Importing fixed-column files

Unfortunately, fixed-column files are more common than I prefer. They are a pain, since you need to specify each column length. I have found this with files with a “.dat” file extension.

We can use the *infix* command to read in fixed-column files. Let’s look at our fixed-column width files.

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
type "dentists7.txt"

```

```

Y. Don Uflossmore 7.2501
Olive Tu'Drill   10.2511
Isaac O'Yerbreath32.7511
Ruth Canaale     22.0011
Mike Avity       8.5000

```

Our name variable is between 1 and 17, years are between 18 and 22. Our fulltime and recommend binaries are only length of 1 with 23 and 24.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
infix str name 1-17 years 18-22 fulltime 23 recom 24 using "dentists7.txt", clear
list
```

(5 observations read)

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

This is a real pain, but luckily Stata has dictionary files to help if one of these files are available. They have a .dct file extension. We already have a fixed-column dictionary with dentists1.dct and we can use it to open a fixed-column width file dentists7.txt

Let's look at our dictionary file.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
type "dentists1.dct"
```

```
infix dictionary {
    str name 1-17 years 18-22 fulltime 23 recom 24
}
```

Let's look at our fixed-column width file

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
type "dentists7.txt"
```

```
Y. Don Uflossmore 7.2501
Olive Tu'Drill    10.2511
Isaac O'Yerbreath32.7511
Ruth Canaale      22.0011
Mike Avity        8.5000
```

Let's use the two together to import the data file

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
infix using "dentists1.dct", using(dentists7.txt) clear
list
```

```
infix dictionary {
  str name 1-17 years 18-22 fulltime 23 recom 24
}
(5 observations read)
```

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

We can combine infile command with a dictionary file.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
type "dentists3.dct"
```

```
infile dictionary using dentists7.txt {
  str17 name %17s "Name of dentist"
  years %5f "Years in practice"
  fulltime %1f "Full time?"
  recom %1f "Recommend Quaddent?"
}
```

Notice that the first line of the dictionary file is infile dictionary using dentists7.txt and then the column widths

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
infile using "dentists3.dct", clear
list
```

```
infile dictionary using dentists7.txt {
  str17 name %17s "Name of dentist"
  years %5f "Years in practice"
```

```

    fulltime %1f    "Full time?"
    recom    %1f    "Recommend Quaddent?"
}

```

(5 observations read)

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1
4.	Ruth Canaale	22	1	1
5.	Mike Avity	8.5	0	0

Dictionary .dct files are very helpful when they are available with fixed-column width files.

2.6.5 Subsetting fixed-column width files

We can use the “in” qualifier to subset the number of rows to import with either *infix* or *infile*.

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell/"
infix str name 1-17 years 18-22 fulltime 23 recom 24 using "dentists7.txt" in 1/3, clear
list

infix using "dentists1.dct" in 1/3, using(dentists7.txt) clear
list

```

(3 observations read)

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1

```

infix dictionary {
    str name 1-17 years 18-22 fulltime 23 recom 24
}

```

(3 observations read)

	name	years	fulltime	recom
1.	Y. Don Uflossmore	7.25	0	1
2.	Olive Tu'Drill	10.25	1	1
3.	Isaac O'Yerbreath	32.75	1	1

2.6.6 2.8 Common errors when reading and importing files

Tip: Don't forget to clear your data. Either the line before or add it as an option , clear

Error 1 If you do not clear your data with the use command, then you will get the message “No; dataset in memory has changed since last saved” - Use *clear* in the line before ingesting the data, or use *clear* as an option.

Error 2 If you do not clear your data with the import, infix, or infile commands, then you will get the message: “You must start with an empty dataset”

2.6.7 Entering data directly into Stata Data Editor

My opinion: Please do not do this. If you need to manually enter data, I don't recommend manually entering data into STATA. I recommend using a spreadsheet, then import the spreadsheet.

If for some reason you don't have access to Excel or a spreadsheet application like OSX Numbers or Google Sheets, then read pages 43-50.

Chapter 3

Saving and Exporting Data

Mitchell Chapter 3

3.1 Saving Stata files

It can be helpful to convert a csv, excel, or other type of file to a Stata dta file. It is also good to consider older types of Stata files Older versions of Stata help with sharing data. (One of my issues with Stata is that newer data files are not compatible with older versions of Stata).

Let's import a comma-delimited file.

```
import delimited using "jul25pub.csv", clear
```

We can use the save and saveold commands to convert this comma-delimited file into a Stata dta file. Don't forget to replace or else you will run into an error when trying to replicate the code.

The *save* command will save data into a Stata version *N* dta file. (Stata 14 for me)

```
save jul25pub, replace
```

If you need to collaborate with someone with an older version of stata, then use *saveold* command. For example, if I want to someone with Stata 12 to use my file, I can use *saveold* and specify version 12 in the options.

```
saveold jul25pub_v12, version(12) replace
```

We can subset as well using the *keep* or *drop* commands.

```
keep if gestfip==24  
save jan25pub, replace  
saveold jan25pub_v12, version(12) replace
```

There is a compress command, but harddrive are large enough today that it should not be a problem.

3.2 Exporting Excel Files

Let's say we want to collaborate with someone that doesn't have Stata or wants to make graphs in Excel. We can export to excel either ".xls" or ".xlsx" from Stata with the export excel command

Let's use our main file again

```
use dentlab, clear
```

We can use the export excel command Don't forget to use replace option and set the firstrow as variable names

```
export excel using "dentlab.xlsx", replace firstrow(variables)
```

Using the command above, if there are value labels, then labels are exported into the binary 0/1 for parttime and recommend. We can use the nolabel option.

```
export excel using "dentlab.xlsx", replace firstrow(variables) nolabel
```

But what we want a "labeled" and an "unlabeled" sheet in the same excel file? The line of code above will replace the file we saved originally. What we can use is the sheet option and create a excel sheet called "Labeled".

```
export excel using "dentlab.xlsx", sheet("Labeled") firstrow(variables) replace
```

Now, let's add another excel sheet called "Unlabeled". This will create a new sheet or replace an existing sheet called "Unlabeled". However, it won't replace the entire excel file. We use the replace option in the sheet subcommand option.

```
export excel using "dentlab.xlsx", sheet("Unlabeled", replace) firstrow(variables) nolabel
```

3.3 Exporting SAS, SPSS, and dBase files

My recommendation is to export csv files instead of SAS and SPSS files. CSV files can be universally read.

Use the *export sasxport8* command to export a SAS XPORT Version 8 file.

```
export sasxport8 jan25pub.v8xpt
```

Use the *export dbase* command to export a dbase file.

```
export dbase jan25pub.dbf
```

Use the *export spss* command to export a SPSS file.

```
export spss jan25pub.sav
```

3.4 Exporting comma and tab delimited files

Our main focus will be exporting CSV files. I recommend when collaborating with others, especially if they are using different software like SAS, R, Python, etc. to use csv. They are universally read, which makes collaboration easier. You may lose labels (or values), but If everyone is not using Stata, then export csv (or excel) is a good idea. The default is comma-delimited.

Our simplest export is:

```
use dentlab
list
export delimited using "dentists_comma.csv", replace
```

This will export the labels instead of the values. If you want to export the values instead of the labels, use the *nolabels* option.

```
export delimited using "dentists_comma.csv", nolabel replace
type "dentists_comma.csv"
```

You also have the option of putting quotes around strings, which can be a good idea if you strings have “,” in them. Otherwise, the csv will think the “,” is a place to delimit and your dataset will be incorrectly exported. The option is simply *quote*.

```
export delimited using "dentists_comma.csv", nolabel quote replace
```

If for some reason you do not want to export the variables names, you can use the novarnames option.

```
export delimited using "dentists_comma.csv", nolabel quote novarnames replace
type "dentists_comma.csv"
```

Sometimes we may need to format our data for exporting. Note that the years for Mike Avity we just replace exported as 8.9300003. To fix this, we can use the format command to set the total width to 5 and the decimal places to 2 with %5.2f

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use dentlab
replace years = 8.93 if name == "Mike Avity"
export delimited using "dentists_comma.csv", nolabel quote replace
type "dentists_comma.csv"
```

```
(1 real change made)
```

```
file dentists_comma.csv saved
```

```
name,years,fulltime,recom
"Y. Don Uflossmore",7.25,0,1
"Olive Tu'Drill",10.25,1,1
"Isaac O'Yerbreath",32.75,1,1
"Ruth Canaale",22,1,1
"Mike Avity",8.9300003,0,0
```

Use the option fmtdata in later versions of Stata

```
format years %5.2f
export delimited using "dentists_comma.csv", nolabel quote replace fmtdata
type "dentists_comma.csv"
```

For more options see:

```
help export delimited
```

3.5 Exporting space-separated files

Don't do this. Use comma-delimited files. If you are really interested in this, please review pages 65-66

3.6 Creating Excel Reports

I liked this addition to the exporting data section. A lot of time, senior leadership may need a report, and with *export excel* (or *putexcel*) we can generate automated reports. The key here is automation, where the pipeline from raw data to reports is done completely through a .do file.

A lot of times it is a pain to get the data analysis to your write-ups, and *export excel* and *putexcel* are great tools to make it easy. We'll cover *estout* to get regression outputs into Word or LaTeX a bit later. *I have used *putexcel* in the past, but we'll cover two ways to fill out Excel reports

We are going to create a Full-time Part-time Report in Excel.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use "dentlab.dta", clear
keep name years fulltime
export excel using "dentrpt1.xlsx", cell(A4) replace
```

file dentrpt1.xlsx saved

	A	B	C	D
2				
3				
4	Y. Don Uflossmore	7.25	part time	
5	Olive Tu'Drill	10.25	full time	
6	Isaac O'Yerbreath	32.75	full time	
7	Ruth Canaale	22	full time	
8	Mike Avity	8.5	part time	
9				

Figure 3.1: Dental Report

Notice this is just exporting our data into an excel file without a shell, but we do place our data starting on line A4 and we do not export variable names.

We'll use the shell report called "dentrpt1-skeleton.xlsx", and we'll fill it out using Stata. This is just an example, but it can be easily applicable in many office settings. It is preferable to fill out the report this way, since if there is a mistake, then it is easy to fix without generating a new report from scratch.

	A	B	C
1	Report #1: Tenure and Full vs. Part Time Status		
2			
3	Dentist Name	Years of Experience	Full vs. Part Time
4			
5			

First we'll use the copy command to copy the Report Shell over our newly created file called "dentrpt1.xlsx".

```
copy "dentrpt1-skeleton.xlsx" "dentrpt1.xlsx", replace
```

Now we will use the "modify" option to modify our report shell to fill out the report with data.

```
export excel "dentrpt1.xlsx", sheet("dentists", modify) cell(A4)
```

	A	B	C
1	Report #1: Tenure and Full vs. Part Time Status		
2			
3	Dentist Name	Years of Experience	Full vs. Part Time
4	Y. Don Uflossmore	7.25	part time
5	Olive Tu'Drill	10.25	full time
6	Isaac O'Yerbreath	32.75	full time
7	Ruth Canaale	22	full time
8	Mike Avity	8.5	part time
9			
10			

Figure 3.2: Dental Report v3

Notice that there are two sheets in the report shell: "dentists" and "Sheet2". We modified the sheet called "dentists" and pasted our data in this sheet starting at cell A4.

We can format our formats, too. We have our report shell in "dentrpt2-skeleton.xlsx". We will do is similar to the first report, but we will add the keepcellfmt option.

```
copy "dentrpt2-skeleton.xlsx" "dentrpt2.xlsx", replace
export excel "dentrpt2.xlsx", sheet("dentists", modify) cell(A4) keepcellfmt
```

Note: this keepcellfmt is unavailable in Stata 14, but you should have it in your newer version.

We'll do something similar, but we will include averages for years and full-time vs part time.

```
copy "dentrpt3-skeleton.xlsx" "dentrpt3.xlsx", replace
export excel "dentrpt3.xlsx", sheet("dentists", modify) cell(A6)
```

	A	B	C
1	Report #3 Tenure and Full vs. Part Time Status		
2	Average Years of Experience	99.9	
3	Percentage Full Time	99.9%	
4			
5	Dentist Name	Years of Experience	Full vs. Part Time
6	<i>Y. Don Uflossmore</i>	7.25	<u>part time</u>
7	<i>Olive Tu'Drill</i>	10.25	<u>full time</u>
8	<i>Isaac O'Yerbreath</i>	32.75	<u>full time</u>
9	<i>Ruth Canaale</i>	22	<u>full time</u>
10	<i>Mike Avity</i>	8.5	<u>part time</u>
11	AAA	99.0000	<u>BB</u>
12			

Figure 3.3: Dental Report v5

You have two options for the percentages: 1) add average formula in Excel or 2) use put excel. It is easy to add formulas in excel, but what if the number of observation changes over time? Then, you would need to modify the Excel sheet. Using put excel you can update the averages without messing with formulas in excel.

It is poor practice to run the summarize command and manually entering it into your report. Use local macros and do it dynamically!

We will set our dentists report to be modified by Stata, and then run the summarize command and take the average stored after the command and put it into excel.

```
putexcel set "dentrpt3.xlsx", sheet("dentists") modify
```

After setting the sheet to be modified, we can write expression, returned results, formulas, graphs, scalars, matrices, etc.

Let's look at what the summarize command stores after being run.

```
summarize years
return list
```

Variable	Obs	Mean	Std. Dev.	Min	Max
years	5	16.15	10.98095	7.25	32.75

scalars:

```

r(N) = 5
r(sum_w) = 5
r(mean) = 16.15
r(Var) = 120.58125
r(sd) = 10.98094941250528
r(min) = 7.25
r(max) = 32.75
r(sum) = 80.75

```

There are several scalars that are stored, such as number of observations $r(N)$, mean $r(\text{mean})$, standard deviation $r(\text{sd})$, etc. If we use the detail option, we'll see more scalars.

We can retrieve those scalars with local macros, such as `'r(mean)'`.

```

summarize years
putexcel B2 = `r(mean)'
summarize fulltime
putexcel B3 = `r(mean)'
putexcel save //Not in older version

```

	A	B	C
1	Report #3 Tenure and Full vs. Part Time Status		
2	Average Years of Experience	16.2	
3	Percentage Full Time	60.0%	
4			
5	Dentist Name	Years of Experience	Full vs. Part Time
6	<i>Y. Don Uflossmore</i>	7.25	<u>part time</u>
7	<i>Olive Tu'Drill</i>	10.25	<u>full time</u>
8	<i>Isaac O'Yerbreath</i>	32.75	<u>full time</u>
9	<i>Ruth Canaale</i>	22	<u>full time</u>
10	<i>Mike Avity</i>	8.5	<u>part time</u>
11	AAA	99.0000	BB
12			

Figure 3.4: Dental Report v6

More on Putexcel <https://blog.stata.com/2017/01/10/creating-excel-tables-with-putexcel-part-1-introduction-and-formatting/>

Chapter 4

Cleaning and Checking Data

Mitchell Chapter 4

Good note from Mitchell at the beginning: Garbage In; Garbage Out.

Before doing analysis, you really need to check your data to look for data quality issues.

4.1 Checking individual variables

We start off with looking for problems in individual variables. Mitchell uses Working Women's Survey (WWS) dataset, but we'll use Current Population Survey (CPS) data, as well.

```
use wws, clear
```

The *describe* command is a useful command to see an overview of your data with variables labels, storage type, and value labels (if any)

```
quietly cd "/Users/Sam/Desktop/Data/CPS/"  
use smalljul25pub  
describe
```

Contains data from smalljul25pub.dta

```
obs:      122,252  
vars:      50  
size:      8,313,136
```

12 Aug 2025 19:17

variable name	storage type	display format	value label	variable label
hrhhid2	int	%8.0g		
hrmis	byte	%8.0g		HRMIS
hrmonth	byte	%8.0g		HRMONTH
hryear4	int	%8.0g		HRYEAR4
gestfips	byte	%8.0g		
gediv	byte	%8.0g		
pulineno	byte	%8.0g		PULINENO
pudis	byte	%8.0g		PUDIS
puiodp1	byte	%8.0g		PUIODP1
puiodp2	byte	%8.0g		PUIODP2
puiodp3	byte	%8.0g		PUIODP3
perrp	byte	%8.0g		
pesex	byte	%8.0g		
pemarit1	byte	%8.0g		
pehspnon	byte	%8.0g		
peeduca	byte	%8.0g		
peafever	byte	%8.0g		
ptdtrace	byte	%8.0g		
prpertyp	byte	%8.0g		
prtage	byte	%8.0g		
pehractt	int	%8.0g		
pehract1	byte	%8.0g		
pehract2	byte	%8.0g		
pemlr	byte	%8.0g		
peio1icd	int	%8.0g		
peio1cow	byte	%8.0g		
peio2cow	byte	%8.0g		
prmjind1	byte	%8.0g		
prmjind2	byte	%8.0g		
prmjocc1	byte	%8.0g		
prmjocc2	byte	%8.0g		
prdtind1	byte	%8.0g		
prdtind2	byte	%8.0g		
prdtocc1	byte	%8.0g		
prdtocc2	byte	%8.0g		
prsjmj	byte	%8.0g		
peernhry	byte	%8.0g		
peernhro	byte	%8.0g		
peernlab	byte	%8.0g		
prerelg	byte	%8.0g		
prdisflg	byte	%8.0g		
pecert1	byte	%8.0g		

```

pecert2      byte      %8.0g
pecert3      byte      %8.0g
pwcmpwgt     long      %12.0g
pternwa      long      %12.0g
ptio1ocd     int       %8.0g
hrhhid       double    %10.0g
ptwk         byte      %8.0g
gtmetsta     byte      %8.0g

```

Sorted by:

You will noticed that we lack value labels and variable labels. Luckily we have a data dictionary on Census's website. Let's label the variables and use the *describe* command again.

```

label variable gestfips "State"
label variable gediv "Census Region"
label variable pternwa "Weekly earnings"
label variable peio1cow "Class of worker in Main Job"
label variable peio1cow "Class of worker in Second Job"
label variable pemlr "Labor Force Status"
describe

```

Contains data from smalljul25pub.dta

```

obs:      122,252
vars:      50
size:      8,313,136

```

12 Aug 2025 19:17

variable name	storage type	display format	value label	variable label
hrhhid2	int	%8.0g		
hrmis	byte	%8.0g		HRMIS
hrmonth	byte	%8.0g		HRMONTH
hryear4	int	%8.0g		HRYEAR4
gestfips	byte	%8.0g		State
gediv	byte	%8.0g		Census Region
pulineno	byte	%8.0g		PULINENO
pudis	byte	%8.0g		PUDIS
puiodp1	byte	%8.0g		PUIODP1
puiodp2	byte	%8.0g		PUIODP2
puiodp3	byte	%8.0g		PUIODP3
perrp	byte	%8.0g		
pesex	byte	%8.0g		
pemaritl	byte	%8.0g		

pehspnon	byte	%8.0g	
peeduca	byte	%8.0g	
peafever	byte	%8.0g	
ptdtrace	byte	%8.0g	
prpertyp	byte	%8.0g	
prtage	byte	%8.0g	
pehractt	int	%8.0g	
pehract1	byte	%8.0g	
pehract2	byte	%8.0g	
pemlr	byte	%8.0g	Labor Force Status
peio1icd	int	%8.0g	
peio1cow	byte	%8.0g	Class of worker in Second Job
peio2cow	byte	%8.0g	
prmjind1	byte	%8.0g	
prmjind2	byte	%8.0g	
prmjocc1	byte	%8.0g	
prmjocc2	byte	%8.0g	
prdtind1	byte	%8.0g	
prdtind2	byte	%8.0g	
prdtocc1	byte	%8.0g	
prdtocc2	byte	%8.0g	
prsjmj	byte	%8.0g	
peernhry	byte	%8.0g	
peernhro	byte	%8.0g	
peernlab	byte	%8.0g	
prerelg	byte	%8.0g	
prdisflg	byte	%8.0g	
pecert1	byte	%8.0g	
pecert2	byte	%8.0g	
pecert3	byte	%8.0g	
pwcmpwgt	long	%12.0g	
pternwa	long	%12.0g	Weekly earnings
ptio1ocd	int	%8.0g	
hrhhid	double	%10.0g	
ptwk	byte	%8.0g	
gtmetsta	byte	%8.0g	

Sorted by:

4.2 One way tabulation

The *tabulation* or *tab* command is very useful to inspect categorical variables.

The *tabulation* or *tab* command is very useful to inspect categorical variables. Let's look at *collgrad*, which is a binary for college graduate status in *wws* data,

and let's look at race data, which is a categorical variable. Let's include the missing option to make sure no observations are missing data

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use wws, clear
tabulate collgrad, missing
tabulate race, missing
```

(Working Women Survey)

college		Freq.	Percent	Cum.
graduate				
0		1,713	76.27	76.27
1		533	23.73	100.00
Total		2,246	100.00	
race		Freq.	Percent	Cum.
1		1,636	72.84	72.84
2		583	25.96	98.80
3		26	1.16	99.96
4		1	0.04	100.00
Total		2,246	100.00	

Race should only be coded between 1 and 3, so we have one miscoded observation. Let's find the observation's idcode

```
list idcode race if race==4
```

(Working Women Survey)

```

+-----+
| idcode  race |
+-----+
2013. |    543     4 |
+-----+
```

4.3 Summarize

The *summarize* or *sum* command is very useful to inspect continuous variables. Let's look at the amount of unemployment insurance in unempins. The values range between 0 and 300 dollars for unemployed insurance received last week.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use wws, clear
summarize unempins
```

(Working Women Survey)

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
unempins	2,246	30.50401	73.16682	0	299

Let's look at weekly earnings in CPS.

```
quietly cd "/Users/Sam/Desktop/Data/CPS/"
use smalljul25pub.dta, replace
gen earnings=pternwa/100
summarize earnings if prerelg==1
```

(28,617 missing values generated)

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
earnings	9,635	1408.951	1251.25	0	6930.64

The data range between \$0 and \$6,930.64 in weekly earnings last week with a mean of 1,408.95 dollars and a standard deviation of 1,251.25 dollars. The max seems a bit high. Let's add the detail option to get more information.

```
summarize earnings if prerelg==1, detail
```

earnings				

	Percentiles	Smallest		
1%	88	0		
5%	250	0		
10%	384	0	Obs	9,635
25%	680	0	Sum of Wgt.	9,635
50%	1060		Mean	1408.951
		Largest	Std. Dev.	1251.25
75%	1730	6930.64		
90%	2700	6930.64	Variance	1565626
95%	3460	6930.64	Skewness	2.688217
99%	6930.64	6930.64	Kurtosis	11.9109

We can see that the mean is \$1,408.95, while the median is \$1,060.00. The mean appears to be skewed rightward, where the skewness is 2.7 and the kurtosis is 11.9. The 95th percentile is 3,460.00 while the 99th percentile is \$6,930.64.

Let's look at wage data in the wws data set

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use wws, clear
summarize wage, detail
```

(Working Women Survey)

hourly wage				

	Percentiles	Smallest		
1%	1.892108	0		
5%	2.801002	1.004952		
10%	3.220612	1.032247	Obs	2,246
25%	4.259257	1.151368	Sum of Wgt.	2,246
50%	6.276297		Mean	288.2885
		Largest	Std. Dev.	9595.692
75%	9.661837	40.19808		
90%	12.77777	40.74659	Variance	9.21e+07
95%	16.73912	250000	Skewness	35.45839
99%	38.70926	380000	Kurtosis	1297.042

For our wws data, our mean of 288.29 dollars/hour appears to be highly skewed rightward and heavily influenced by outliers. Our median hourly wage is 6.7 dollars per hour and our 99th percentile hourly wage is 38.7 dollars per hour, but our mean is 288.29 dollar/hour. Our skewness is 35, which means our wage data are highly skewed. Our Kurtosis shows that the max outliers are heavily skewing the results. A normal distributed variable should have a kurtosis of 3, and our kurtosis is 1297. When we see the 2 largest observations, they are 250,000 and 380,000 dollars/hour.

This means we have found potential data measurement issues. Let's look at the outliers

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
list idcode wage if wage > 1000
```

```
| idcode      wage |
|-----|
```

```

893. |    3145    380000 |
1241. |    2341    250000 |
      +-----+

```

Let's look at ages, which should range from 21 to 50 years old. We can use both the *summarize* and *tabulate* commands. *Tabulate* can be very useful with continuous variables if the range is not too large.

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
summarize age
tabulate age

```

Variable	Obs	Mean	Std. Dev.	Min	Max
age	2,246	36.25111	5.437983	21	83

age in current year	Freq.	Percent	Cum.
21	2	0.09	0.09
22	8	0.36	0.45
23	16	0.71	1.16
24	35	1.56	2.72
25	44	1.96	4.67
26	50	2.23	6.90
27	50	2.23	9.13
28	63	2.80	11.93
29	49	2.18	14.11
30	55	2.45	16.56
31	57	2.54	19.10
32	60	2.67	21.77
33	56	2.49	24.27
34	95	4.23	28.50
35	213	9.48	37.98
36	224	9.97	47.95
37	171	7.61	55.57
38	175	7.79	63.36
39	167	7.44	70.79
40	139	6.19	76.98
41	148	6.59	83.57
42	111	4.94	88.51
43	110	4.90	93.41
44	98	4.36	97.77

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45		45	2.00	99.78
46		1	0.04	99.82
47		1	0.04	99.87
48		1	0.04	99.91
54		1	0.04	99.96
83		1	0.04	100.00
-----+				
Total		2,246	100.00	

Let's look at the outliers

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
list idcode age if age > 50
```

```
      | idcode  age |
      |-----|
2205. |    80   54 |
2219. |    51   83 |
      |-----+
```

We have again found potential data measurement problems. The data is supposed to range from 21 to 50, but there are two observations with 54 and 83. It is possible that they were incorrectly entered, and they are supposed to be 45 and 38, respectively.

4.4 Crosstabulation - categorical by categorical variables

The two-way tabulation through the *tabulate* command is a very useful way to look for data quality problems or to double check binaries created. Let's look at variables *metro*, which is a binary for whether or not the observation lives in a metropolitan areas, and *ccity*, which is whether or not the observation lives in the center of the city.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabulate metro ccity, missing
```

```
Does woman |
live in | Does woman live in
metro | city center?
```

area?	0	1	Total
-----+-----			
0	665	0	665
1	926	655	1,581
-----+-----			
Total	1,591	655	2,246

An alternative is to count the number of observations that shouldn't be there. So we'll count the number of observations not in a metro area but in a city center. I personally don't use it, but it has its uses.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
count if metro==0 & ccity==1
```

0

Let's look at married and nevermarried. There should be no individuals that appear in both married and nevermarried.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabulate married nevermarried, missing
```

	Woman never been		
	married		
married	0	1	Total
-----+-----+-----			
0	570	234	804
1	1,440	2	1,442
-----+-----+-----			
Total	2,010	236	2,246

There may be observation that have been married, but not currently married. There should be no observations that are both married and never married.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
count if married == 1 & nevermarried == 1
list idcode married nevermarried if married == 1 & nevermarried == 1
```

2

4.4. CROSSTABULATION - CATEGORICAL BY CATEGORICAL VARIABLES 51

	idcode	married	neverm-d
1523.	1758	1	1
2231.	22	1	1

We have found another potential data measurement issue. There are 2 observations that are married and never married.

Let's look at college graduates and years of school completed. We can use *tabulate* or the *table* commands. I personally prefer *tabulate*, since we can look for missing.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabulate yrschool collgrad, missing
```

Years of school completed	college graduate 0	1	Total
8	69	1	70
9	55	0	55
10	84	0	84
11	123	0	123
12	943	0	943
13	174	2	176
14	180	7	187
15	81	11	92
16	0	252	252
17	0	106	106
18	0	154	154
.	4	0	4
Total	1,713	533	2,246

We can use the *table* command

```
table yrschool collgrad
```

Years of school completed	college graduate 0	1
---------------------------------	--------------------------	---

```

-----+-----
      8 |    69    1
      9 |    55
     10 |    84
     11 |   123
     12 |   943
     13 |   174    2
     14 |   180    7
     15 |    81   11
     16 |       252
     17 |      106
     18 |      154
-----

```

Now let's find the ID with potential problems

```
list idcode if yrschool < 16 & collgrad == 1
```

```

      | idcode |
      |-----|
195. |  4721 |
369. |  4334 |
464. |  4131 |
553. |  3929 |
690. |  3589 |
      |-----|
1092. |  2681 |
1098. |  2674 |
1114. |  2640 |
1124. |  2613 |
1221. |  2384 |
      |-----|
1493. |  1810 |
1829. |   993 |
1843. |   972 |
1972. |   689 |
2114. |   312 |
      |-----|
2174. |   172 |
2198. |   107 |
2215. |    63 |
2228. |    25 |
2229. |    24 |
      |-----|
2230. |    23 |
      +-----+

```

There is at least one potential measurement problem. There is an observation with 8 years of schooling completed listed as a college graduate. We also see that there are 2 observations with 13 years and 7 with 14 years. These may be two-year college graduates, and these observations may not be problematic.

4.5 Checking categorical by continuous variables

It is also helpful to look at continuous variables by different categories. Let's look at the binary variable for union, if the observation is a union member or not by `uniondues`.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
summarize uniondues, detail
```

Union Dues paid last week					

	Percentiles	Smallest			
1%	0	0			
5%	0	0			
10%	0	0	Obs		2,242
25%	0	0	Sum of Wgt.		2,242
50%	0		Mean		5.603479
		Largest	Std. Dev.		9.029045
75%	10	29			
90%	22	29	Variance		81.52365
95%	26	29	Skewness		1.35268
99%	29	29	Kurtosis		3.339635

The median union dues is 0 dollars, which makes sense. The mean is 5.6 dollars, while the maximum value is 29 dollars.

Let's use the `bysort` command, which will sort our data by the categorical variable specified.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
bysort union: summarize uniondues
```

```
-> union = 0
```

Variable	Obs	Mean	Std. Dev.	Min	Max
----------	-----	------	-----------	-----	-----

```
-----+-----
uniondues |      1,413      .094126      1.502237          0          27
```

```
-> union = 1
```

```
Variable |      Obs      Mean      Std. Dev.      Min      Max
-----+-----
uniondues |      461     14.65944     8.707759          0          29
```

```
-> union = .
```

```
Variable |      Obs      Mean      Std. Dev.      Min      Max
-----+-----
uniondues |      368     15.41304     8.815582          0          29
```

The mean for not in a union is less than 1 dollar with 1,413 observations The mean for union is about 15 dollars with 461 observations The mean for missing is about 15 dollars with 368 observations

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabulate uniondues if union==0, missing
```

```
Union Dues |
paid last |
week |      Freq.      Percent      Cum.
-----+-----
      0 |      1,407      99.29      99.29
     10 |           1       0.07      99.36
     17 |           1       0.07      99.44
     26 |           2       0.14      99.58
     27 |           2       0.14      99.72
      . |           4       0.28     100.00
-----+-----
    Total |      1,417     100.00
```

We see about 4 observations have union dues. We may or may not need to recode them to 0. It is possible that someone may be not be a union members, but still have to pay an agency fee. Let's assume that they are not in 14(b) state..

The recode command can to create a new variable if a person pay union dues. Now let's compare union members to those who pay union dues

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
recode uniondues (0=0) (1/max=1), generate(paysdues)
tabulate union paysdues, missing
```

(784 differences between uniondues and paysdues)

RECODE of uniondues (Union Dues paid last week)				
union	paid last week		.	Total
worker	0	1		
-----+	-----	-----	-----	-----
0	1,407	6	4	1,417
1	17	444	0	461
.	7	361	0	368
-----+	-----	-----	-----	-----
Total	1,431	811	4	2,246

Let's list the non-union members paying union dues. Note: we use the `abb(#)` option to abbreviate the observation to 20 characters

```
list idcode union uniondues if union==0 & (uniondues > 0) & !missing(uniondues), abb(20)
```

	idcode	union	uniondues

561.	3905	0	10
582.	3848	0	26
736.	3464	0	17
1158.	2541	0	27
1668.	1411	0	27

2100.	345	0	26

Let's look at another great command for observing categorical and continuous variables together: `tabstat`

We could use `tab` with a `sum` option

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tab married, sum(marriedyrs)
```

| Summary of Years married (rounded

		to nearest year)		
married		Mean	Std. Dev.	Freq.
0		0	0	804
1		5.5409154	3.5521383	1,442
Total		3.5574354	3.8933494	2,246

Or, we can use *tabstat* which gives us more options

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabstat marriedyrs, by(married) statistics(n mean sd min max p25 p50 p75) missing
```

Summary for variables: marriedyrs
by categories of: married (married)

married		N	mean	sd	min	max	p25	p50
0		804	0	0	0	0	0	0
1		1442	5.540915	3.552138	0	11	2	6
Total		2246	3.557435	3.893349	0	11	0	2

No observation that said they were never married reported years of marriage

Let's look at current years of experience and everworked binary

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabstat currexp, by(everworked) statistics(n mean sd min max) missing
```

Summary for variables: currexp
by categories of: everworked (Has woman ever worked?)

everworked		N	mean	sd	min	max
0		60	0	0	0	0
1		2171	5.328881	5.042181	0	26
Total		2231	5.185567	5.048073	0	26

Let's total years of experience, which is currexp plus prevexp

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
generate totexp=currexp+prevexp
tabstat totexp, by(everworked) statistics(n mean sd min max) missing
```

(15 missing values generated)

Summary for variables: totexp

by categories of: everworked (Has woman ever worked?)

everworked	N	mean	sd	min	max
0	60	0	0	0	0
1	2171	11.57761	4.552392	1	29
Total	2231	11.26625	4.865816	0	29

Everyone with at least one year experience has experience working

Note: Three ways to 1. bysort x_cat: summarize x_con 2. tab x_cat, sum(x_con) 3. tabstat x_con, by(x_cat) statistics(...)

4.6 Checking continuous by continuous variables

We will compare two continuous variables to continuous variables Let's look at unemployment insured received last week if the hours were greater than 30 and hours are not missing.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
summarize unempins if hours > 30 & !missing(hours)
```

Variable	Obs	Mean	Std. Dev.	Min	Max
unempins	1,800	1.333333	16.04617	0	287

The mean is around 1.3 and our range is between our expected 0 and 287

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
count if (hours>30) & !missing(hours) & (unempins>0) & !missing(unempins)
```

19

Let's look at the hours count

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
tabulate hours if (hours>30) & !missing(hours) & (unempins>0) & !missing(unempins)
```

usual hours worked	Freq.	Percent	Cum.
35	1	5.26	5.26
38	2	10.53	15.79
40	14	73.68	89.47
65	1	5.26	94.74
70	1	5.26	100.00
Total	19	100.00	

It seems that 19 observations worked more than 30 hours and received unemployment insurance. This could be potential data issues, or it is possible that the person got unemployment insurance before working again. It depends on the data collection and data generation process.

Let's look at age and years married to find any unusual observations

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
generate agewhenmarried = age - marriedyrs
summarize agewhenmarried
tabulate agewhenmarried
```

Variable	Obs	Mean	Std. Dev.	Min	Max
agewhenmarried	2,246	32.69368	6.709341	13	73

agewhenmarried	Freq.	Percent	Cum.
ied			

-----+-----			
13	1	0.04	0.04
14	4	0.18	0.22
15	11	0.49	0.71
16	8	0.36	1.07
17	18	0.80	1.87
18	19	0.85	2.72
19	14	0.62	3.34
20	20	0.89	4.23
21	36	1.60	5.83
22	42	1.87	7.70
23	52	2.32	10.02
24	65	2.89	12.91
25	67	2.98	15.89
26	81	3.61	19.50
27	73	3.25	22.75
28	93	4.14	26.89
29	90	4.01	30.90
30	118	5.25	36.15
31	91	4.05	40.20
32	124	5.52	45.73
33	109	4.85	50.58
34	109	4.85	55.43
35	149	6.63	62.07
36	138	6.14	68.21
37	119	5.30	73.51
38	124	5.52	79.03
39	119	5.30	84.33
40	89	3.96	88.29
41	80	3.56	91.85
42	62	2.76	94.61
43	50	2.23	96.84
44	43	1.91	98.75
45	23	1.02	99.78
46	2	0.09	99.87
47	1	0.04	99.91
48	1	0.04	99.96
73	1	0.04	100.00
-----+-----			
Total	2,246	100.00	

There was one person who was 73. However, the dataset is for women 21 to 50, so this observation may have been accidentally entered incorrectly. The correct value may be 37.

Let's look for anyone under 18. Some states allow under 18 marriages, but it

still is suspicious.

```
tabulate agewhenmarried if agewhenmarried < 18
```

agewhenmarr ied	Freq.	Percent	Cum.
13	1	2.38	2.38
14	4	9.52	11.90
15	11	26.19	38.10
16	8	19.05	57.14
17	18	42.86	100.00
Total	42	100.00	

Let's use the same strategy for years of experience deducted from age to find the first age working.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
generate agewhenstartwork = age - (prevexp + currexp)
tabulate agewhenstartwork if agewhenstartwork < 16
```

(15 missing values generated)

agewhenstar twork	Freq.	Percent	Cum.
8	1	1.49	1.49
9	1	1.49	2.99
12	1	1.49	4.48
14	20	29.85	34.33
15	44	65.67	100.00
Total	67	100.00	

Some states allow work at the age of 15, but anything below looks suspicious and would need to be investigated for potential data issues.

We can also look for the number of age of children. We should expect that the age of the third child is never older than the second child

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
table kidage2 kidage3 if numkids==3
count if (kidage3 > kidage2) & (numkids==3) & !missing(kidage3)
count if (kidage2 > kidage1) & (numkids>=2) & !missing(kidage2)

```

Age of second child		0	1	Age of third child						
				2	3	4	5	6	7	
0		12								
1		10	9							
2		11	8	10						
3		10	12	6	8					
4		10	12	10	7	5				
5		12	11	9	3	6	8			
6		9	8	10	6	5	6	6		
7		7	6	7	9	4	14	12	6	
8			5	11	7	6	14	6	11	
9				8	13	10	7	12	9	
10					15	3	10	6	12	
11						9	8	3	13	
12							16	9	6	
13								11	5	
14									8	

0

0

Now let's check the age of the observation when the first child was born

```

quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
quietly generate agewhenfirstkid = age - kidage1
tabulate agewhenfirstkid if agewhenfirstkid < 18

```

agewhenfirs tkid		Freq.	Percent	Cum.
3		1	0.51	0.51
5		2	1.01	1.52
7		2	1.01	2.53

8		5	2.53	5.05
9		8	4.04	9.09
10		7	3.54	12.63
11		10	5.05	17.68
12		10	5.05	22.73
13		20	10.10	32.83
14		30	15.15	47.98
15		27	13.64	61.62
16		39	19.70	81.31
17		37	18.69	100.00
Total		198	100.00	

There are some suspicious observations after this tabulation. This would require investigation into data quality issues.

Scatter plots can be helpful as well

```
twoway scatter hours wage if hours > 0 & wage > 0 & wage < 1000
graph export "/Users/Sam/Desktop/Econ 645/Stata/hour_wage.png", width(500) replace
```

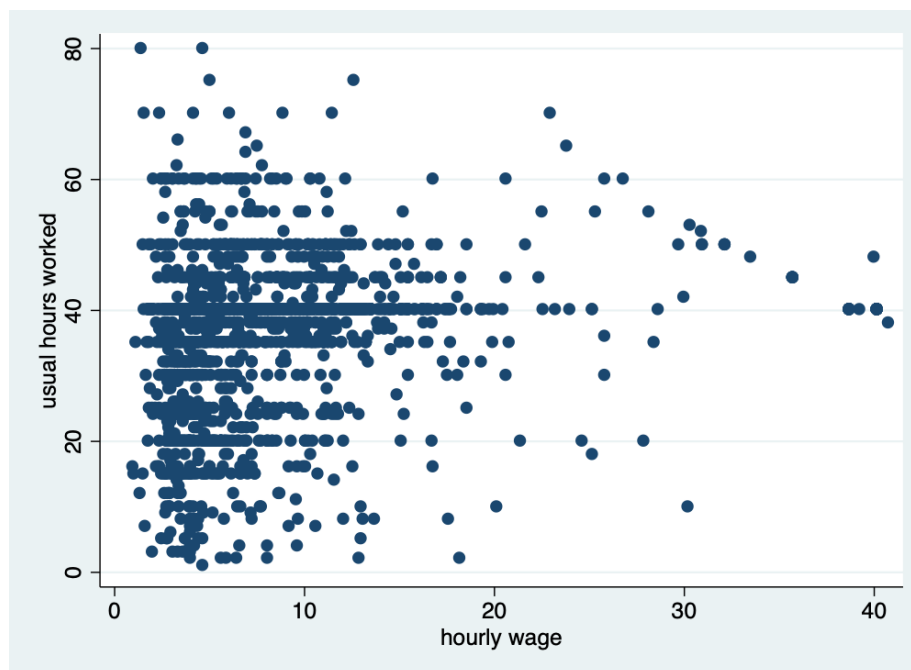


Figure 4.1: Hours and Wages

4.7 Fixing and correcting errors in data

Let go ahead and fix some of these errors that we have found. A word of caution is that you may need to talk to the data stewards about the best ways to correct the data. You don't want to attempt to fix the measurement error and introduce additional problems. Institutional knowledge of the data is very helpful before correcting errors.

Let's fix when race was equal to 4 when there are only 3 categories. Let's add a note for future users, which is good practice for replication.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
list idcode race if race==4
replace race=1 if idcode == 543
note race: race changed to 1 (from 4) for idcode 543
tabulate race
```

```
      | idcode   race |
      |-----|
2013. |    543      4 |
      +-----+
```

(1 real change made)

race	Freq.	Percent	Cum.
1	1,637	72.89	72.89
2	583	25.96	98.84
3	26	1.16	100.00
Total	2,246	100.00	

Let's fix college graduate when there were only 8 years of education.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
list idcode collgrad if yrschool < 12 & collgrad == 1
replace collgrad = 0 if idcode == 107

note collgrad: collgrad changed to 0 (from 1) for idcode 107
tab yrschool collgrad
```

```
      | idcode   collgrad |
      |-----|
```

```
2198. |      107      1 |
      +-----+
```

(1 real change made)

Years of school completed	college graduate 0	1	Total
8	70	0	70
9	55	0	55
10	84	0	84
11	123	0	123
12	943	0	943
13	174	2	176
14	180	7	187
15	81	11	92
16	0	252	252
17	0	106	106
18	0	154	154
Total	1,710	532	2,242

Let's fix age which where the digits were switched, and we will make a note of it.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use wws, clear
list idcode age if age > 50
replace age = 38 if idcode == 51
replace age = 45 if idcode == 80
note age: the value of 83 was corrected to 38 for idcode 51
note age: the value of 54 was corrected to 45 for idcode 80
tab age
```

```
      | idcode  age |
      +-----+
2205. |      80    54 |
2219. |      51    83 |
      +-----+
```

(1 real change made)

(1 real change made)

age in current year	Freq.	Percent	Cum.
21	2	0.09	0.09
22	8	0.36	0.45
23	16	0.71	1.16
24	35	1.56	2.72
25	44	1.96	4.67
26	50	2.23	6.90
27	50	2.23	9.13
28	63	2.80	11.93
29	49	2.18	14.11
30	55	2.45	16.56
31	57	2.54	19.10
32	60	2.67	21.77
33	56	2.49	24.27
34	95	4.23	28.50
35	213	9.48	37.98
36	224	9.97	47.95
37	171	7.61	55.57
38	176	7.84	63.40
39	167	7.44	70.84
40	139	6.19	77.03
41	148	6.59	83.62
42	111	4.94	88.56
43	110	4.90	93.46
44	98	4.36	97.82
45	46	2.05	99.87
46	1	0.04	99.91
47	1	0.04	99.96
48	1	0.04	100.00
Total	2,246	100.00	

Let's look at our notes with the *note* command.

note

_dta:

1. This is a hypothetical dataset and should not be used for analysis purposes

age:

1. the value of 83 was corrected to 38 for idcode 51
2. the value of 54 was corrected to 45 for idcode 80

race:

1. race changed to 1 (from 4) for idcode 543

collgrad:

1. collgrad changed to 0 (from 1) for idcode 107

4.8 Identifying duplicates

The `duplicates` command can be helpful, but it will only find duplicates that are exactly the same **unless you specify which variables with duplicates list vars**. We'll cover this a bit later.

We need to be careful with duplicate observations. When working with panel data we will want multiple observations of the same cross-sectional unit, but there should be multiple observations for different time units and not the same time unit.

We can start with our `duplicates` list command to find rows that are completely the same. This command will show all the duplicates. **Note:** for duplicate records to be found, all data needs to be the same.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use "dentists_dups.dta", clear
duplicates list
```

Duplicates in terms of all variables

group:	obs:	name	years	fulltime	recom
1	4	Mike Avity	8.5	0	0
1	6	Mike Avity	8.5	0	0
1	8	Mike Avity	8.5	0	0
2	1	Olive Tu'Drill	10.25	1	1
2	11	Olive Tu'Drill	10.25	1	1
3	2	Ruth Canaale	22	1	1
3	3	Ruth Canaale	22	1	1

We'll have a more condensed version with `duplicates` examples. It will give an example of the duplicate records for each group of duplicate records.

`duplicates` examples

Duplicates in terms of all variables

group:	#	e.g. obs	name	years	fulltime	recom
1	3	4	Mike Avity	8.5	0	0
2	2	1	Olive Tu'Drill	10.25	1	1
3	2	2	Ruth Canaale	22	1	1

We can find the a consice list of duplicate records with `duplicates` report.

`duplicates` report

Duplicates in terms of all variables

copies	observations	surplus
1	4	0
2	4	2
3	3	2

The `duplicates` tag command can return the number of duplicates for each group of duplicates (note the separator option puts a line in the table after `#` rows).

```
duplicates tag, generate(dup)
list, separator(2)
```

Duplicates in terms of all variables

	name	years	fulltime	recom	dup
1.	Olive Tu'Drill	10.25	1	1	1
2.	Ruth Canaale	22	1	1	1
3.	Ruth Canaale	22	1	1	1
4.	Mike Avity	8.5	0	0	2

5.		Mary Smith	3	1	1	0	
6.		Mike Avity	8.5	0	0	2	

7.		Y. Don Uflossmore	7.25	0	1	0	
8.		Mike Avity	8.5	0	0	2	

9.		Mary Smith	27	0	0	0	
10.		Isaac O'Yerbreath	32.75	1	1	0	

11.		Olive Tu'Drill	10.25	1	1	1	
+-----+							

Let's sort our data for better legibility, and then we will add lines in our output table whenever there is a new group.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use "dentists_dups.dta", clear
sort name years
list, sepby(name years)
```

		name	years	fulltime	recom	

1.		Isaac O'Yerbreath	32.75	1	1	

2.		Mary Smith	3	1	1	

3.		Mary Smith	27	0	0	

4.		Mike Avity	8.5	0	0	
5.		Mike Avity	8.5	0	0	
6.		Mike Avity	8.5	0	0	

7.		Olive Tu'Drill	10.25	1	1	
8.		Olive Tu'Drill	10.25	1	1	

9.		Ruth Canaale	22	1	1	
10.		Ruth Canaale	22	1	1	

11.		Y. Don Uflossmore	7.25	0	1	
+-----+						

Notice! There are two observations for Mary Smith, but one has 3 years of experience and the other one has 27 years, so it does not appear as a duplicate.

If there were too many variables, which could use our data browser.

```
browse if dup > 0
```

We will want to drop our duplicates, but not the first observation. The `duplicates drop` command will drop the duplicate records but still keep one observation of the duplicate group.

```
duplicates drop
```

Let's use a new dataset that is a bit more practical

```
use "wws.dta", replace
```

Let's check for duplicate idcodes with the *isid* command. If there is a duplicate idcode, the command will return an error.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use "wws.dta", replace
isid idcode
```

(Working Women Survey)

Or just use `duplicates list` for the variable of interest

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
quietly use "wws.dta", replace
duplicates list idcode
```

Duplicates in terms of idcode

(0 observations are duplicates)

There are no duplicate idcodes, so we should expect that `duplicates list` will not return any duplicates, since every single values needs to be the same for there to be a duplicate record.

Let's use a dataset that does have duplicates. Let's check for duplicate idcodes.

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use "wws_dups.dta", clear
isid idcode
```

```
variable idcode does not uniquely identify the observations
r(459);
```

```
r(459);
```

Or

```
quietly cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use "wvs_dups.dta", clear
duplicates list idcode, sepby(idcode)
```

Duplicates in terms of idcode

```
+-----+
| group:  obs:  idcode |
+-----+
|      1   1088   2831 |
|      1   2248   2831 |
+-----+
|      2   1244   3905 |
|      2   1245   3905 |
+-----+
|      3    277   4214 |
|      3   2247   4214 |
+-----+
```

I prefer *duplicates list var1*, since an error will break the do file.

Let's generate our dup variable for identifying the number of duplicates per group of duplicates.

```
duplicates tag idcode, generate(iddup)
```

Let's generate a different duplicate variable to find complete duplicate records. Notice that we do not specify a variable after tag.

```
duplicates tag, generate(alldup)
```

Let's list the data

```
list idcode age race yrschool occupation wage if iddup==1 & alldup==0
```

Duplicates in terms of idcode

Duplicates in terms of all variables

```
+-----+
| idcode  age  race  yrschool  occupa~n      wage |
```

1244.	3905 36 1 14 11 4.339774	
1245.	3905 41 1 10 5 7.004828	

It seems that idcode 3905 is a different person with a possible incorrect idcode. Let's fix that.

```
replace idcode=5160 if idcode==3905 & age==41
list idcode age race yrschool occupation wage if iddup==1 & alldup==0
```

(1 real change made)

	idcode age race yrschool occupa~n wage	

1244.	3905 36 1 14 11 4.339774	
1245.	5160 41 1 10 5 7.004828	

Let's look observation that are complete duplicates.

```
duplicates report
list idcode age race yrschool occupation wage if iddup==1 & alldup==1
```

Duplicates in terms of all variables

	copies observations surplus	

	1 2244 0	
	2 4 2	

	idcode age race yrschool occupa~n wage	

277.	4214 35 1 17 13 11.27214	
1088.	2831 37 2 8 8 2.697261	
2247.	4214 35 1 17 13 11.27214	
2248.	2831 37 2 8 8 2.697261	

We need to keep 1 observation of the duplicates. Use the *duplicates drop* command.

```
duplicates drop
duplicates report
```

Duplicates in terms of all variables

(2 observations deleted)

Duplicates in terms of all variables

```
-----
      copies | observations      surplus
-----+-----
           1 |          2246           0
-----
```

4.9 Practice

Let's work the Census CPS.

Pull the Census CPS. - You can download the file and import it, or - You can use Stata to pull the file.

```
import delimited using "https://www2.census.gov/programs-surveys/cps/datasets/2025/bas
```

Let's save a copy of the csv file in Stata

Let's grab the data dictionary. Hint search: "Census CPS 2025 data dictionary"

Let's check our data. - What variable contains our laborforce status? - What variable contains our weekly earnings? - Does everyone employed have weekly earnings? Hint: Try *summarize* - What flag with our weekly earning that provides information about available data? - Try *tabstat* with the flag and weekly earnings. What do you find? - Try *tabstat* with statistics(n mean sd p25 p50 p75) - What are our identifier variable(s)? How many did you find? - Are there any duplicate records?

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

UCLA Advanced Research Computing. (2025). *Stata Learning Modules*. Retrieved from <https://stats.oarc.ucla.edu/stata/modules/>